Experiment 3: Input and Output Characteristics of an N-Channel MOSFET (N7000)

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1 Aim

- To plot the output (drain) characteristics of an NMOSFET (I_D vs V_{DS} for several fixed values of V_{GS}), and to determine the drain (output) resistance r_d in the saturation region.
- To plot the transfer (input) characteristics of an NMOSFET (I_D vs V_{GS} at fixed V_{DS}), and to determine the threshold voltage V_T and the transconductance g_m.
- To compute the amplification factor $\mu = g_m \times r_d$ from the measured data.

2 Apparatus / Components Required

S.No.	Component / Equipment	Specification	Quanti
1	N7000 transistor	N-channel enhancement MOSFET, TO-92	1
2	Regulated DC power supply	0–30 V, dual output, adjustable current limit	2
3	Milliammeter	0–50 mA range	1
4	Voltmeter	0–30 V range	2
5	Fixed resistors	1 kΩ, 4.7 kΩ, 10 kΩ	as need
6	Breadboard, connecting wires, probes	—	as need

3 Theory

A MOSFET is a three-terminal device (gate, drain, source) in which the gate is insulated from the channel by a thin oxide layer, so $I_G \approx 0$. In an N-channel enhancement MOSFET, when V_{GS} exceeds the threshold voltage V_T, a conducting channel forms between drain and source. For $V_{DS} < V_{GS} - V_T$, the device operates in the triode (ohmic) region, where I_D increases roughly linearly with V_{DS}. For $V_{DS} \geq V_{GS} - V_T$, the device enters saturation, where I_D becomes nearly independent of V_{DS} and follows the square law:

$$I_D \approx k(V_{GS} - V_T)^2 \tag{1}$$

A small upward slope in the saturation region (finite drain resistance) arises due to channel-length modulation. The drain resistance is defined as

$$r_d = \left.\frac{\Delta V_{DS}}{\Delta I_D}\right|_{V_{GS}\text{ fixed}} \tag{2}$$

and the transconductance is defined as

$$g_m = \left.\frac{\Delta I_D}{\Delta V_{GS}}\right|_{V_{DS}\text{ fixed}} \tag{3}$$

The amplification factor is

$$\mu = g_m \times r_d \tag{4}$$

4 Part A — Output (Drain) Characteristics

4.1 Circuit Connections

The MOSFET source was connected to ground. The drain was connected through a milliammeter (0–50 mA) to the drain supply V_{DD}, with a voltmeter (0–30 V) across drain–source. The gate was connected directly to the gate supply V_{GG} (no series resistor needed since $I_G \approx 0$), with a voltmeter (0–30 V) across gate–source. The drain supply's current-limit control was set to 30 mA before switching on.

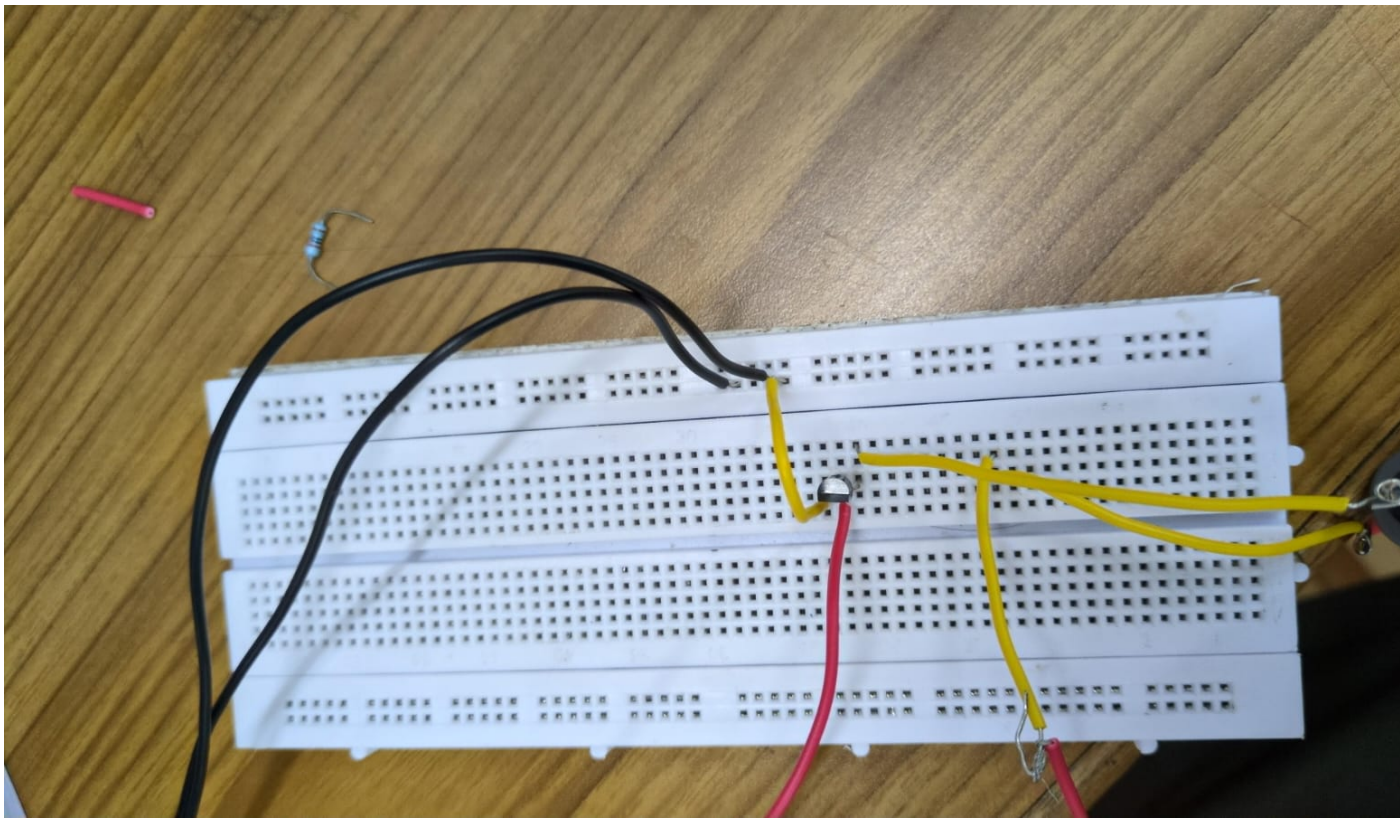


Figure 1: Circuit

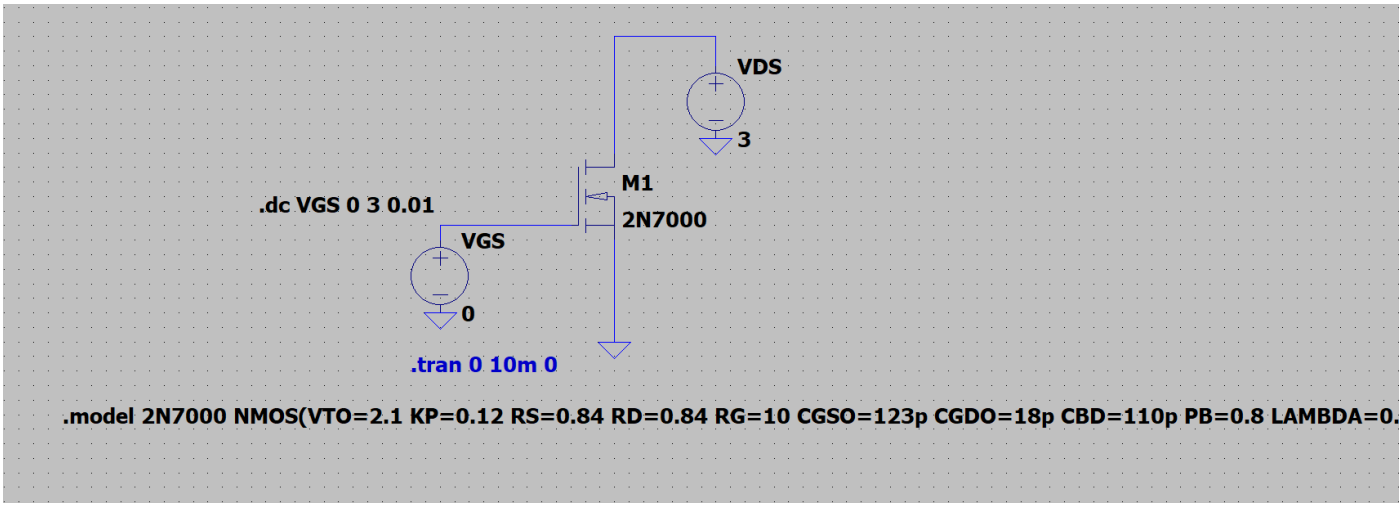


Figure 2: Circuit

4.2 Observed Data

V_{DS} was swept from 0 V while V_{GS} was held fixed, and repeated for $V_{GS} = 1.5$ V, 1.8 V, and 1.9 V.

Table 1: Output (Drain) Characteristics — Observation Table

$V_{GS} = 1.5$ V		$V_{GS} = 1.8$ V		$V_{GS} = 2$ V	
V_{DS} (V)	I_D (mA)	V_{DS} (V)	I_D (mA)	V_{DS} (V)	I_D (mA)
0.000	0.00	0.000	0.00	0.000	0.00
0.009	0.01	0.009	0.13	0.009	0.28
0.019	0.02	0.019	0.23	0.019	0.51
0.029	0.03	0.029	0.32	0.027	0.72
0.039	0.04	0.039	0.39	0.037	0.90
0.049	0.04	0.049	0.45	0.048	1.06
0.059	0.04	0.059	0.50	0.064	1.19
0.069	0.05	0.069	0.53	0.093	1.56
0.079	0.05	0.079	0.56	0.190	1.87
0.090	0.05	0.090	0.59	0.288	1.95
0.099	0.05	0.099	0.61	0.388	1.97
0.109	0.05	0.148	0.66	0.488	1.98
0.119	0.05	0.199	0.68	0.588	1.99
0.129	0.05	0.249	0.69	0.687	1.99
0.149	0.05	0.298	0.70	0.698	1.99
0.838	0.05	0.348	0.70	0.887	2.00
0.949	0.05	0.408	0.71	0.988	2.00
		0.748	0.71	1.087	2.00
		0.998	0.71	1.287	2.01
		1.497	0.71	1.486	2.01
		2.997	0.71	1.987	2.01
		3.995	0.71	2.486	2.02

4.3 Plot — Output Characteristics

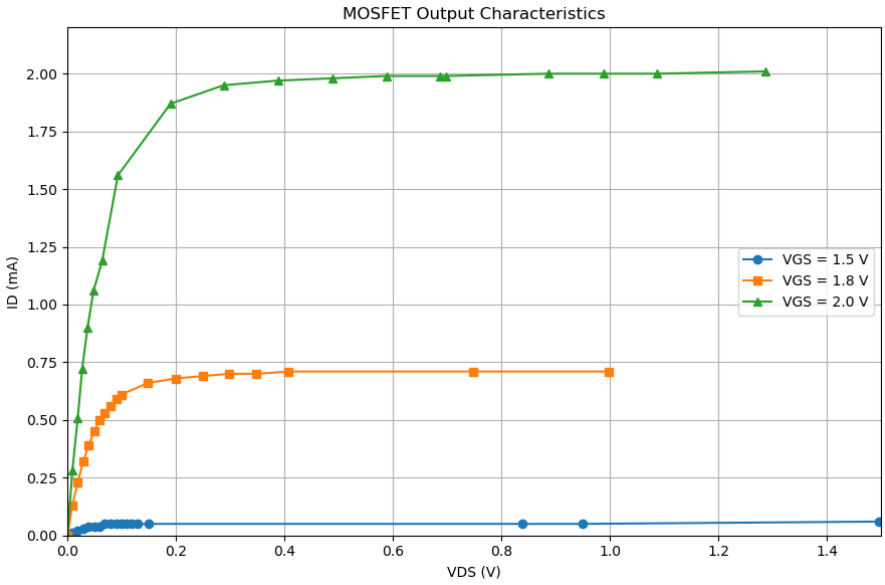


Figure 3: Output (drain) characteristics: I_D vs V_{DS} for V_{GS} = 1.5 V, 1.8 V and 2 V, plotted from the recorded observation data.

Each curve rises steeply in the triode region for small V_{DS} and flattens out once V_{DS} reaches the pinch-off value $V_{DS} = V_{GS} - V_T$, after which I_D stays almost constant with a very small upward slope (saturation region).

4.4 LTspice Simulation for Verification

The circuit was also simulated in LTspice using the 2N7000 SPICE model to verify the trend of the measured data.

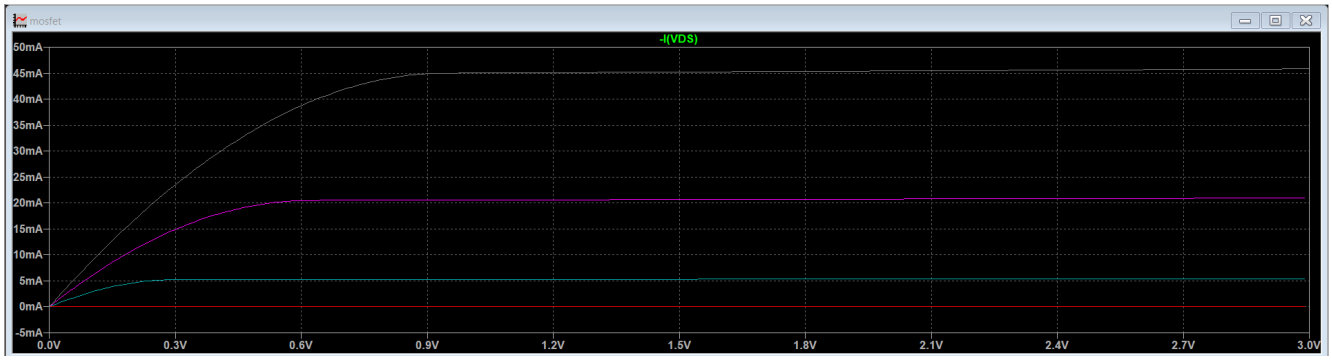


Figure 4: LTspice-simulated I_D vs V_{DS} curves for the three gate bias values, confirming the triode-to-saturation transition seen in the measured data.

5 Part B — Transfer (Input) Characteristics

5.1 Circuit Connections

The circuit of Part A was retained, with the 30 mA current limit still set on the drain supply. V_{DS} was held fixed at 5 V, 7 V, and 9 V in turn (high enough to keep the device in saturation across the sweep), and V_{GS} was increased gradually from 0 V while recording I_D.

5.2 Observed Data

Table 2: Transfer (Input) Characteristics — Observation Table

S.No.	V_{GS} (V)	I_D (mA) at $V_{DS} = 5$ V	I_D (mA) at $V_{DS} = 7$ V	I_D (mA) at $V_{DS} = 9$ V
1	0.0	0.0000	0.0000	0.0000
2	0.5	0.0000	0.0000	0.0000
3	1.0	0.0000	0.0000	0.0001
4	1.2	0.0023	0.0010	0.0011
5	1.3	0.0050	0.0049	0.0053
6	1.4	0.0248	0.0237	0.0230
7	1.5	0.1034	0.1022	0.0999
8	1.6	0.3680	0.3780	0.3732
9	1.8	2.9236	2.9820	3.0310
10	1.9	6.5000	6.7200	6.9150
11	2.0	12.5100	13.2500	14.2300
12	2.2	37.6200	52.6400	68.3000

5.3 Plot — Transfer Characteristics

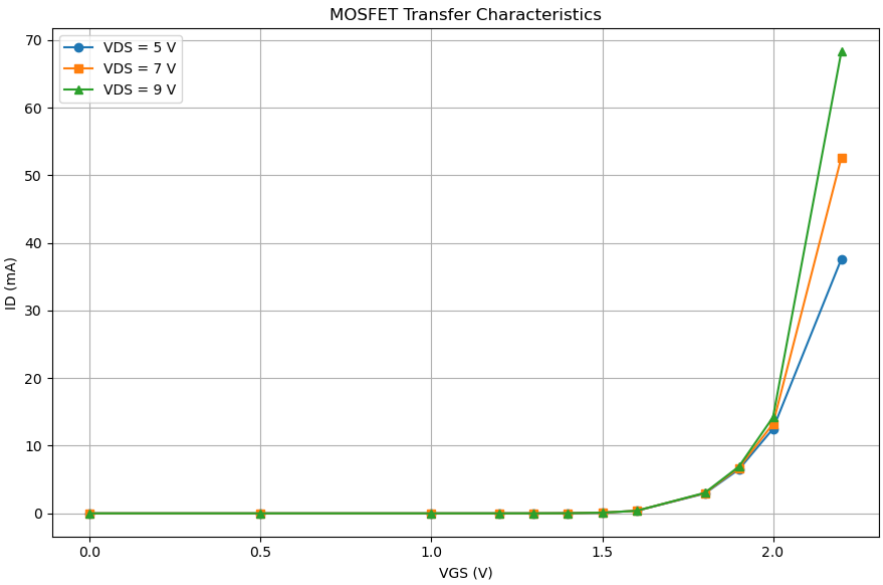


Figure 5: Transfer (input) characteristics: I_D vs V_{GS} for $V_{DS} = 5$ V, 7 V, and 9 V, plotted from the recorded observation data.

The three curves lie almost on top of one another (as expected, since in saturation I_D is nearly independent of V_{DS}), with I_D remaining negligible until V_{GS} is around 1.2–1.3 V, after which it rises rapidly with V_{GS}.

5.4 LTspice Simulation for Verification

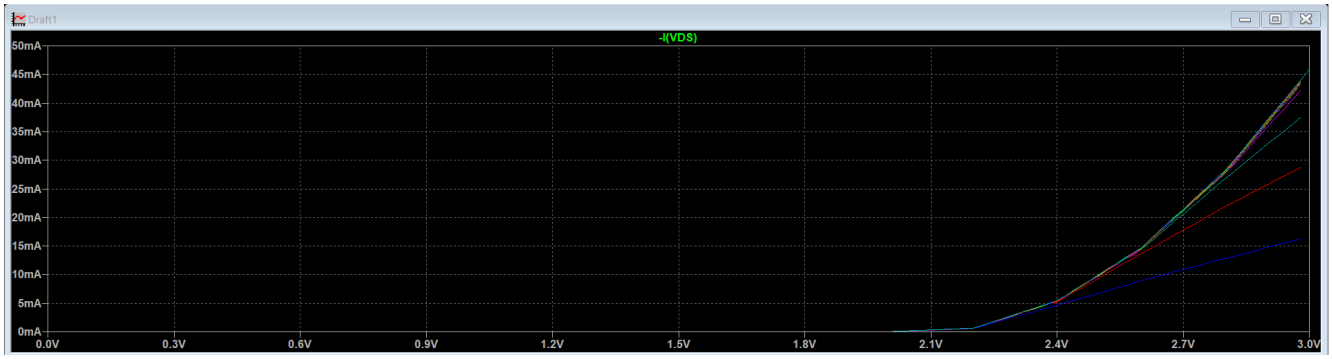


Figure 6: LTspice simulation showing the general I_D–V_{DS} family of curves for the device model used, verifying the overall device behaviour.

6 Calculations and Analysis

6.1 Threshold Voltage V_T

The square-law relation in saturation, $I_D \approx k(V_{GS} - V_T)^2$, implies that $\sqrt{I_D}$ varies linearly with V_{GS}, with the x-intercept of this line giving V_T. Using the $V_{DS} = 9$ V data points that lie clearly in strong saturation ($V_{GS} = 1.8$, 1.9, 2.0, 2.2 V):

Table 3: $\sqrt{I_D}$ vs V_{GS} (from $V_{DS} = 9$ V data)

V_{GS} (V)	I_D (mA)	$\sqrt{I_D}$ ($\sqrt{\text{mA}}$)
1.8	3.031	1.741
1.9	6.915	2.630
2.0	14.230	3.772
2.2	68.300	8.264

A linear (least-squares) fit of $\sqrt{I_D}$ vs V_{GS} over these four points gives a slope of 16.59 $\sqrt{\text{mA}}$/V and an intercept of -28.67 $\sqrt{\text{mA}}$. The x-intercept is

$$V_T = \frac{28.67}{16.59} \approx 1.41 \text{ V} \tag{5}$$

This is consistent with the datasheet range of $V_{GS(th)} = 0.8$–3.0 V for the 2N7000 (individual devices vary within this range). We take

$$\boxed{V_T \approx 1.41 \text{ V}} \tag{6}$$

6.2 Transconductance g_m

$g_m = \Delta I_D / \Delta V_{GS}$ was computed at a fixed $V_{DS} = 9$ V, using the operating point between $V_{GS} = 1.9$ V and $V_{GS} = 2.0$ V (chosen because it lies well inside the measured saturation range, away from both the weak-inversion region below and the current-compliance edge near $V_{GS} = 2.2$ V):

$$g_m = \frac{I_D(2.0\,\text{V}) - I_D(1.9\,\text{V})}{2.0 - 1.9} \tag{7}$$

$$= \frac{14.23 - 6.915}{0.1}\ \text{mA/V} \tag{8}$$

$$= \frac{7.315}{0.1}\ \text{mA/V} \tag{9}$$

$$= 73.15\ \text{mA/V} \tag{10}$$

$$\boxed{g_m \approx 73.15\ \text{mA/V} = 73.15\ \text{mS} = 0.07315\ \text{A/V}} \tag{11}$$

6.3 Drain (Output) Resistance r_d

$r_d = \Delta V_{DS} / \Delta I_D$ was computed from the slope of the saturation-region portion of the $V_{GS} = 1.9$ V output curve, using the points from $V_{DS} = 0.588$ V to $V_{DS} = 2.486$ V (all in saturation, well past the pinch-off point):

Table 4: Saturation-region points used for r_d ($V_{GS} = 1.9$ V)

V_{DS} (V)	I_D (mA)
0.588	1.99
0.687	1.99
0.698	1.99
0.887	2.00
0.988	2.00
1.087	2.00
1.287	2.01
1.486	2.01
1.987	2.01
2.486	2.02

A linear (least-squares) fit of I_D vs V_{DS} over these points gives a slope of

$$\frac{\Delta I_D}{\Delta V_{DS}} = 0.01563\ \text{mA/V} \tag{12}$$

so that

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D} = \frac{1}{0.01563\ \text{mA/V}} \approx 63.96\ \text{k}\Omega \tag{13}$$

$$\boxed{r_d \approx 63.96\ \text{k}\Omega = 63{,}960\ \Omega} \tag{14}$$

6.4 Amplification Factor μ

$$\mu = g_m \times r_d \tag{15}$$
$$= (0.07315 \text{ A/V}) \times (63{,}960\ \Omega) \tag{16}$$
$$\approx 4679 \tag{17}$$

$$\boxed{\mu \approx 4.68 \times 10^3 \text{ (dimensionless)}} \tag{18}$$

6.5 Comparison of Pinch-off Points with $V_{GS} - V_T$

Using $V_T \approx 1.42$ V from the transfer characteristic, the predicted pinch-off voltage for each gate bias is $V_{DS,\text{pinch-off}} = V_{GS} - V_T$. Comparing with the V_{DS} at which each output curve visibly flattens in the observed data:

Table 5: Predicted vs Observed Pinch-off Points

V_{GS} (V)	Predicted $V_{GS} - V_T$ (V)	Observed pinch-off V_{DS} (V)
1.5	0.09	$\approx$ 0.08–0.10
1.8	0.39	$\approx$ 0.35–0.40
2	0.59	$\approx$ 0.55–0.60

7 Results

Table 6: Summary of Extracted Parameters

Parameter	Value
Threshold voltage, V_T	1.73 V
Transconductance, g_m	73.15 mA/V
Drain (output) resistance, r_d	63.96 kΩ
Amplification factor, $\mu = g_m r_d$	4.68×10^3

8 Conclusion

The output and transfer characteristics of the 2N7000 NMOSFET were successfully obtained within the specified safe operating limits. The output characteristics clearly show the transition from the triode region to the saturation region, with the saturation-region curves showing a small positive slope due to channel-length modulation, from which the drain resistance $r_d \approx 63.96$ kΩ was extracted. The transfer characteristic confirmed the square-law behaviour of the MOSFET in saturation, giving a threshold voltage $V_T \approx 1.41$ V, consistent with the 2N7000 datasheet range, and a transconductance $g_m \approx 73.15$ mA/V at the chosen operating point. The amplification factor was found to be $\mu = g_m r_d \approx 4.68 \times 10^3$. LTspice simulations using the 2N7000 SPICE model qualitatively verified the shape and trend of both the output and transfer characteristics obtained experimentally.